ic_autolab round 3 (generic track): report

Protocol: PROTOCOL.md, frozen 2026-11-04T02:44:38Z at git `9a3b129f9dde3557e84c784a5bed33c6fceb048c`; route `local`. Counted GAE only (deterministic, host independent); no wall-time claim; development loop, no validation-gate row. Loop home: `../`; ledger: `../ledger.json`.

Outcome

k	class	sel	sel / inc at w = 0 [95%]	w = 0.1	w = 1	band	S sel (w = 0 / 0.1 / 1)	S inc	S rho-neg	distance sel / rho
20	**no_challenger**	none (best candidate `ksum4-n10`, train S 10.285 / 10.593 / 13.365)	-	-	-	-	-	8.699 / 9.385 / 15.556	1.694 / 1.821 / 2.957	inc: 5.1x / 5.2x / 5.3x
24	**no_challenger**	none (best candidate `rfmitm2-n81`, train S 12.170 / 13.209 / 22.560)	-	-	-	-	-	6.575 / 7.022 / 11.048	1.242 / 1.354 / 2.366	inc: 5.3x / 5.2x / 4.7x

Classes follow the protocol's decision rule. Distance to rho is the distance remaining, not a gate. The reasons per size:

- k = 20: no_challenger
- k = 24: no_challenger

Train (development screen, in sample)

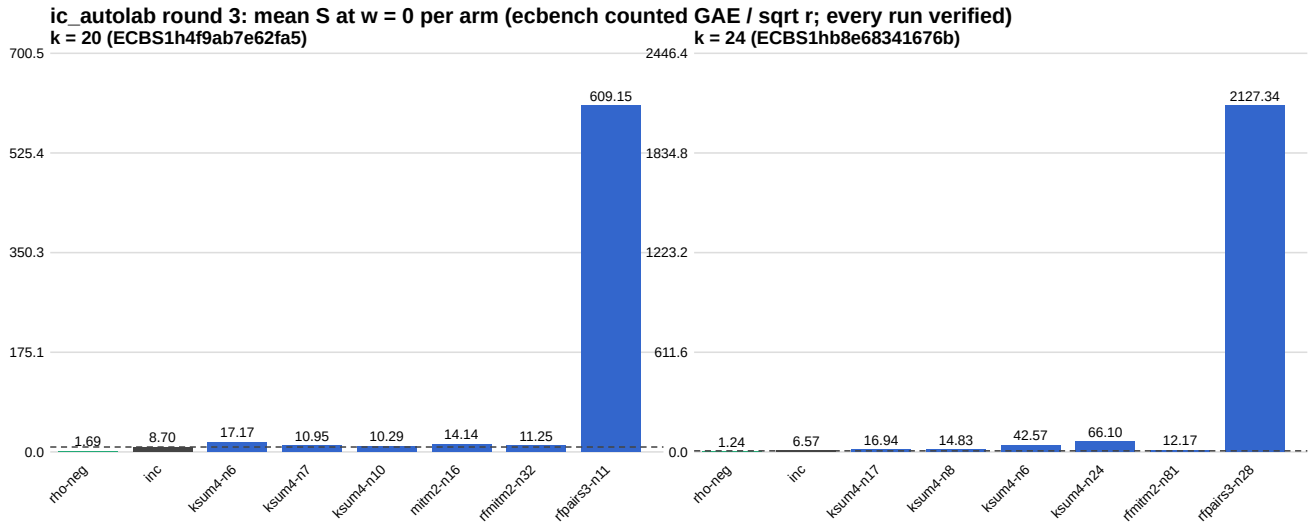
k = 20: session `ECBS1h4f9ab7e62fa5` (run exit 0, verify exit 0, receipt sha256 `ba8e6ce658ee3bc38902ede86463b3a78d0fdbf908d2faf9a1c6e265493404f7`)

arm	family	N	factor	base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
`rho-neg`	rho	-	-		`rho.negation`	16	16	1.694 / 1.821 / 2.957	matched reference
`inc`	ksum3	11	`prime-projected-abscissa:size=11,scan_cap=176`		`ksum:m=3`	16	16	8.699 / 9.385 / 15.556	incumbent
`ksum4-n6`	ksum4	6	`prime-projected-abscissa:size=6,scan_cap=96`		`ksum:m=4`	16	16	17.166 / 19.585 / 41.363	cross-size transfer: ksum4-n12 is the k = 24 incumbent at N / N_a = 0.148
`ksum4-n7`	ksum4	7	`prime-projected-abscissa:size=7,scan_cap=112`		`ksum:m=4`	16	16	10.952 / 12.120 / 22.630	cross-size transfer neighbourhood: N = 1.4 x 5 (ksum4 from k = 24)
`ksum4-n10`	ksum4	10	`prime-projected-abscissa:size=10,scan_cap=160`		`ksum:m=4`	16	16	10.285 / 10.593 / 13.365	cross-size transfer neighbourhood: N = 2 x 5 (ksum4 from k = 24)
`mitm2-n16`	mitm2	16	`prime-projected-abscissa:size=16,scan_cap=256`		`mitm:negation_folded=1`	16	16	14.140 / 15.861 / 31.351	cross-size transfer neighbourhood:

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
							N = 0.5 x 32 (mitm2 from k = 28)
rfmitm2-n32	rfmitm2	32 prime-projected-abscissa:size=32,scan_cap=512	rootfind-mitm:m=2	16	16	11.254 / 12.039 / 19.111	exploration: family rfmitm2 untried at k = 20, N = 1 x N_a
rfpairs3-n11	rfpairs3	11 prime-projected-abscissa:size=11,scan_cap=176	rootfind-pairs:m=3	16	16	609.147 / 665.105 / 1168.730	exploration: family rfpairs3 untried at k = 20, N = 0.35 x N_a

k = 24: session ECBS1hb8e68341676b (run exit 0, verify exit 0, receipt sha256 45b8426ac95e1294bb5952c5bd79f6704ad713061624b332706c717a3fc4e272)

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
rho-neg	rho	- -	rho.negation	16	16	1.242 / 1.354 / 2.366	matched reference
inc	ksum4	12 prime-projected-abscissa:size=12,scan_cap=192	ksum:m=4	16	16	6.575 / 7.022 / 11.048	incumbent
ksum4-n17	ksum4	17 prime-projected-abscissa:size=17,scan_cap=272	ksum:m=4	16	16	16.936 / 17.122 / 18.796	incumbent neighbourhood: N = 1.4 x 12
ksum4-n8	ksum4	8 prime-projected-abscissa:size=8,scan_cap=128	ksum:m=4	16	16	14.828 / 17.388 / 40.424	incumbent neighbourhood: N = 0.7 x 12
ksum4-n6	ksum4	6 prime-projected-abscissa:size=6,scan_cap=96	ksum:m=4	16	16	42.568 / 50.707 / 123.958	incumbent neighbourhood: N = 0.5 x 12
ksum4-n24	ksum4	24 prime-projected-abscissa:size=24,scan_cap=384	ksum:m=4	16	16	66.098 / 66.257 / 67.688	incumbent neighbourhood: N = 2 x 12
rfmitm2-n81	rfmitm2	81 prime-projected-abscissa:size=81,scan_cap=1296	rootfind-mitm:m=2	16	16	12.170 / 13.209 / 22.560	exploration: family rfmitm2 untried at k = 24, N = 1 x N_a
rfpairs3-n28	rfpairs3	28 prime-projected-abscissa:size=28,scan_cap=448	rootfind-pairs:m=3	16	16	2127.339 / 2332.827 / 4182.224	exploration: family rfpairs3 untried at k = 24, N = 0.35 x N_a



grey: incumbent; green: matched rho; blue: candidate (every run verified); red: candidate with an unverified or failed run. Bars are recorded band means; the dashed line is the incumbent.

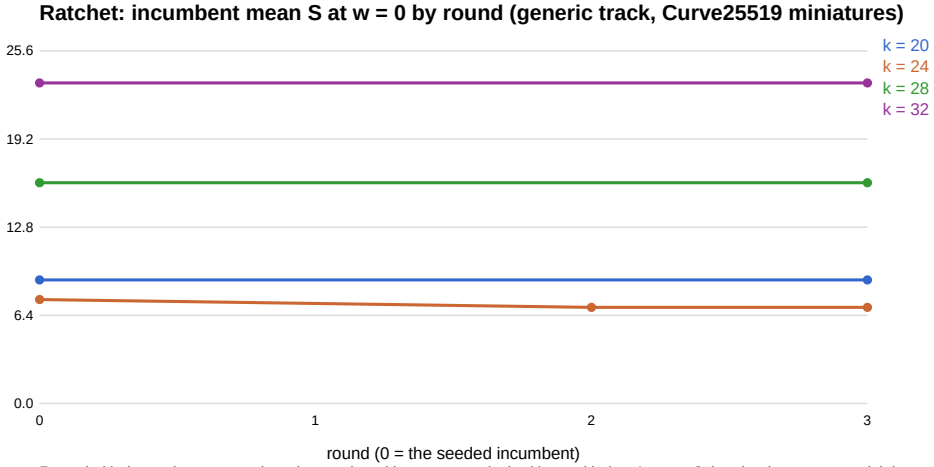
train screen

Selection

- k = 20: no_challenger (Pareto alternative at w = 1: `ksum4-n10`)
- k = 24: no_challenger (Pareto alternative at w = 1: `ksum4-n17`)

Ledger after this round

k	incumbent	since round	S (w = 0 / 0.1 / 1)	distance to rho
20	`ksum3-n11` (`prime-projected-abscissa:size=11,scan_cap=176` + `ksum:m=3`)	0	8.987 / 9.710 / 16.217	5.7x / 5.7x / 6.0x
24	`ksum4-n12` (`prime-projected-abscissa:size=12,scan_cap=192` + `ksum:m=4`)	2	6.985 / 7.514 / 12.271	6.0x / 5.9x / 5.6x
28	`mitm2-n203` (`prime-projected-abscissa:size=203,scan_cap=3248` + `mitm:negation_folded=1`)	0	16.057 / 17.564 / 31.130	16.6x / 16.6x / 16.7x
32	`mitm2-n512` (`prime-projected-abscissa:size=512,scan_cap=8192` + `mitm:negation_folded=1`)	0	23.310 / 25.520 / 45.410	24.7x / 24.6x / 24.3x



Recorded ledger values; a step down is a ratcheted improvement (paired interval below 1 at w = 0, band order at every weight).

ratchet

Evidence and deviations

- ecbench binary sha256: `3fd22853485e388782ac0821295c56f81725fc832fd3610978be8088b5098bca`; `--cpus auto`, allow_busy = true; `verify --replay 8` per session; audit receipts in `analysis/`; sessions in `sessions/` (records in the artifact store when large).

- Route: the local route is an explicitly requested development screen, not the Iso Lab EKS default; counts are deterministic and replayed, wall time is not claimed. No row of this round enters the IC leaderboard or `benchmarks/README.md` (validation gate).
- Train selection is in sample (disclosed); the holdout is fresh (new `target_seed`, excluded exposures). Failed, timed-out and regressing arms are retained above.